

Real-Time Gravitational Wave Detection and Source Characterization with Machine Learning

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Outline

Part I — Background

- Gravitational waves & general relativity
- LIGO detector network
- Compact binary coalescences
- Multi-messenger astronomy
- Traditional methods & their limits

Part II — AFRAME

- ML-based GW detection
- Architecture, training, performance
- Full O3 catalog search

Gravitational Waves

- Predicted by General Relativity (Einstein, 1916)
- Ripples in spacetime — transverse, quadrupolar
- Two polarizations: **plus** (h_+) and **cross** ($h_\times$)

Strain: $\Delta L/L \sim 10^{-21}$

At 4 km scale: displacement $< 10^{-18}$ m
(1/1000 the diameter of a proton)

- First direct detection: **GW150914**, September 2015
- Nobel Prize in Physics 2017 (Weiss, Barish, Thorne)

LIGO Detector Network

- Fabry-Pérot Michelson interferometers
 - Hanford, WA (H1) · Livingston, LA (L1)
 - Arm length: **4 km**
- Also: Virgo (Italy), KAGRA (Japan)

Noise sources:

- *High frequency*: photon shot noise
- *Low frequency*: seismic, gravity gradient
- *Mid-band*: thermal, quantum

O4 sensitivity:

- BBH horizon distance: $\sim 5 \text{ Gpc}$

Compact Binary Coalescences

Three classes of compact binary mergers:

- **BBH** — Binary black holes (most common)
- **BNS** — Binary neutron stars
- **NSBH** — Neutron star–black hole

Three signal phases:

1. *Inspiral* — frequency sweeps up ("chirp")

duration: < 0.2 s (BBH) to minutes (BNS)

2. *Merger* — peak luminosity, nonlinear GR

Multi-Messenger Astronomy

GW170817 — August 17, 2017

First binary neutron star + EM detection

Time after merger	Event
0 s	GW signal (LIGO/Virgo)
+1.7 s	Short gamma-ray burst (Fermi)
+11 hrs	Optical kilonova discovered
+9 days	X-ray afterglow
+16 days	Radio afterglow

Traditional Detection: Matched Filtering

Cross-correlation with template waveforms → SNR time series

Q-transform spectrogram showing the characteristic "chirp"

- Template bank: **~500,000 waveforms** searched simultaneously
- Triggers ranked by SNR; FAR estimated via timeslides
- State-of-the-art sensitivity — but computationally intensive
- Median end-to-end detection latency: **12–41 seconds**

The Speed Problem

Step	Traditional	Goal
Detection	12–41 s	< 5 s
Sky localization	~20 s	< 2 s
Full PE	1–12 hours	< 1 s
Total	30+ min	~6 s

EM counterparts fade in **minutes to hours**.
Every second of latency costs observing time.

Our Approach

Two complementary ML pipelines:

AFRAME: Overview

Algorithm For Real-time Astrophysical Machine learning Events

- Processes **H1 + L1** data simultaneously
- Input: 1.5 s windows of whitened strain at 2048 Hz
- Output: scalar detection statistic (analogous to SNR)
- Inference rate: **4 Hz** (0.25 s stride)
- 1 s top-hat integration → peak finding
→ events

AFRAME: Neural Network Architecture

Modified ResNet34 with 1D convolutions

Component	Choice	Why
Convolutions	1D	Time-series input, not images
Normalization	Group Norm	No train/infer mismatch
Loss	Binary cross-entropy	Detection framing

AFRAME: Training Strategy

Training data:

- Real noise: **4.7 days** coincident O3 data
- Signals: **75,000 BBH** waveforms (IMRPhenomPv2)

Augmentation teaches the network what matters:

Strategy	Label	Teaches
Noise inversion/ reversal	noise	Noise diversity
Signal muting (1 detector)	noise	Coincidence

AFRAME: Augmentation Visualized

Background: noise inversion & time-reversal

Signal: muting, swapping, curriculum SNR schedule

Training: ~**43 hours** on single V100 GPU · Model stays performant for **months** without retraining

AFRAME: Detection Example

GW190828_063405 — neural network output (top) and 1 s integrated detection statistic (bottom). Peak clearly exceeds threshold.

End-to-end detection latency: **3.1 seconds** · Neural network inference: **< 10 ms**

AFRAME: Sensitivity

Metric: Sensitive volume–time (VT)

Higher VT = more of the universe searched

Performance vs matched filtering:

- **State-of-the-art** for $35\text{--}35\text{ }M_{\odot}$ BBH
- **Comparable** to GstLAL, MBTA, PyCBC for $35\text{--}20\text{ }M_{\odot}$ systems
- *Reduced* sensitivity for $m < 20\text{ }M_{\odot}$

AFRAME: Full O3 Catalog Search

Full O3 (Nov 2019 – Mar 2020) · **202.4 days** coincident livetime · **100 years** background via timeslides

38 GWTC-3 candidates with $p_{\text{astro}} > 0.5$ + **3** additional from IAS/OGC catalogs
Individual matched filter pipelines found 25–61 events each

O3 misses (15 total)	Reason
5 events	Below training mass prior
10 events	Low network SNR (< 12.6)

AFRAME detects **high-mass BBH** most confidently — consistent with the sensitivity results

AFRAME: O3 Detection Statistics

Sensitive volume vs FAR — AFRAME competitive with individual matched filter pipelines across the O3 run

Missed/found events by p_{astro} — events above SNR threshold reliably recovered

AMPLFI: What We Need to Characterize

AFRAME detects — but **characterization** determines the EM follow-up strategy

Parameter	Why it matters for follow-up
Sky location (θ, ϕ)	Where to point the telescope
Distance d	Which galaxy to search
Chirp mass M_c	BBH / BNS / NSBH classification
Mass ratio q	Is a neutron star remnant possible?

These parameters must arrive **before** the kilonova fades — minutes, not hours

AMPLFI: The Parameter Estimation Gap

Pipeline	Latency	Method	Limitation
BAYESTAR	~20 s	Analytical	Needs matched filter triggers
LALInference	1–12 hrs	MCMC	Too slow for real-time
Bilby	1–12 hrs	Nested sampling	Too slow for real-time

AMPLFI goal: full posterior $p(\theta \mid d)$ in **< 1 second**

No matched filter triggers · Runs on the same GPU as AFRAME

AMPLFI: Normalizing Flows

Key idea: learn an invertible mapping from simple $\rightarrow$ complex distribution

$$z \sim \mathcal{N}(0, I) \longleftrightarrow \theta \sim p(\theta | d)$$

Why this works:

- Posterior $p(\theta|d)$ is complex, multi-modal
- Conditioned on data summary $\rightarrow$ adapts each event
- Once trained, sampling is

instantaneous

AMPLFI: Architecture

Two-module design:

1. Embedding network (self-supervised pre-training)

- Input: whitened strain + PSD (H1, L1, $\pm V1$)
- Compresses data to fixed-dim latent summary
- Marginalizes over time-of-arrival
(merger time provided by AFRAME)

2. Normalizing flow (coupling layers)

- Conditioned on data embedding

AMPLFI: GPU-Accelerated Training

On-GPU, on-the-fly waveform generation:

- IMRPhenomD implemented in **PyTorch**
- GPU whitening, FFTs, PSD estimation
- 1000 waveforms in **~0.15 seconds**
- No pre-generated datasets needed

Training on real detector noise (not Gaussian):

- Captures actual glitches and non-stationarities

- Generalizes across observing runs

AMPLFI: Sky Localization

Searched area: AMPLFI vs BAYESTAR

Searched volume: AMPLFI vs BAYESTAR

Similar 90% credible areas to BAYESTAR · **Tighter searched volume** (better 3D localization)

Validated on 7 GWTC-3 events · **< 1 second** inference vs 20 s (BAYESTAR)

AMPLFI: Parameter Recovery

Chirp mass recovery — injected vs recovered

PP-plot — uniform CDFs confirm well-calibrated posteriors

Chirp mass **unbiased** across BBH parameter space · PP-plot confirms **posterior calibration**

Consistent with Bilby for 7 GWTC-3 events · orders of magnitude faster

Online Deployment: O4c

End-to-end production pipeline deployed
August – November 2025 (LIGO O4c)

**First ML-based GW detection pipeline
deployed in LVK production during a live
run**

Hardware: Single NVIDIA A30 GPU

- AFRAME + AMPLFI co-deployed
- Shared GPU memory — no data transfer bottleneck

Integration:

- GraceDB (LVK alert database)

O4c: Pipeline Architecture

Data flow:

1. Snapshotter caches overlapping 1.5 s windows
2. Whitener — online PSD estimation
3. **AFRAME** — fires trigger on detection
4. **AMPLFI** — receives merger time, produces posterior
5. GraceDB — uploads sky map + classification

O4c: Results

24 CBC candidates with H1-L1 data available during O4c

23/24 detected above any threshold

(96%)

19/24 above public alert threshold

(79%)

9 events where AFRAME was
the preferred pipeline

- 1 non-detection: **power outage** at detector site (not algorithm failure)
- AMPLFI sky maps distributed via GCN for all 9 preferred events
- Duty cycle and FAR requirements met throughout O4c
- Handled glitches and data quality issues in production autonomously

O4c: Example Sky Map

AMPLFI sky map for S251021u — distributed via GCN alert ~6 seconds after merger detection. Localizes the source to enable immediate EM telescope follow-up.

End-to-End Latency

Step	AFRAME/AMPLFI	Traditional
Detection	~3.1 s	12–41 s
Sky localization	< 1 s	~20 s (BAYESTAR)
Full PE	< 1 s	1–12 hours
Alert distribution	~2 s	~2 s
Total	~6 s	30+ min

~5× faster end-to-end than matched filtering
for detection + localization combined

What this enables:

- Alerts before the optical kilonova peaks

Key Contributions

AFRAME

- State-of-the-art ML GW detection
- Competitive with matched filtering for high-mass BBH
- **3.1 s** detection latency
- Full O3 search: **38 GWTC-3** events + **3** new candidates
- First ML pipeline deployed in **LVK production**

AMPLFI

- Sub-second Bayesian PE
- Sky localization competitive

Limitations & Future Work

Current Limitations

AFRAME:

- Reduced sensitivity for $m < 20 M_{\odot}$
(long-duration signals)
- Two-detector only (H1 + L1)
- No BNS/NSBH support yet

AMPLFI:

- Less precise than Bilby per event
- BBH only; BNS/NSBH not yet trained
- Precession, higher modes excluded

Thank You

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AFRAME: github.com/ML4GW/aframe · *AMPLFI:* github.com/ML4GW/amplfi